

Causal State Binding: An Intervention-Based Evaluation Framework for Language Agents

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Abstract

Language agents store memories and goals for later use, yet successful task completion alone does not show whether those records influenced an action. We evaluate causal state binding—the dependence of subsequent action on an agent’s saved records—by changing the records before execution. Controlled action-choice experiments establish consistent responses to supplied information, including advantages over removal, scrambling and matched controls in four open-weight models. Tool-execution experiments then test records written by the acting agent itself. Across 160 tasks per core configuration, removing memory reduces success by 75.0 percentage points in Sol and 54.1 in DeepSeek; removing the saved goal reduces it by 36.6 and 25.9 points. Restoring the record recovers performance, while access to the same facts elsewhere reduces dependence on memory. On 128 additional tasks per API configuration, removing task content while retaining transaction identifiers reduces success by 65.6 points (95% interval 59.4–71.5) in Sol and 56.6 points (52.7–60.5) in DeepSeek. Replacing content redirects actions toward the altered record, separating information use from correctness. As an application, a fixed predictor combining repeated answers and state interventions is evaluated on 800 new software issues across four models. Its area under the receiver-operating-characteristic curve (AUC) is 0.9403, compared with 0.9365 for a generic diagnostic comparator funded to match diagnostic token use; the small gain does not meet the prespecified multiple-comparison criterion. Together, the experiments show how saved task information contributes to action, when another source can replace it, and how changing its content changes what an agent does.

Keywords:

language agents, agent evaluation, state-action binding, causal interventions, reliability, SWE-bench

1. Introduction

Language agents use memories, plans and persistent goals to carry information from one interaction to the next (Yao et al., 2023; Shinn et al., 2023; Park et al., 2023). A recorded budget can constrain a later purchase; a saved scheduling policy can determine

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which feasible appointment to select. The central question is whether the agent actually uses that record when it acts. Task success alone leaves the question open: the agent may obtain the same information from its current instructions or interaction history. To identify the record’s contribution, we compare actions after changing the record while holding the task and execution environment fixed.

We call this relation *causal state binding* (CSB): subsequent action depends on task-relevant information in a saved record. Memories and goals together form the *exposed state*, which can be inspected and changed before execution. Our central finding is that agent-written memories and goals make substantial, recoverable contributions to tool execution in the tested API configurations. Removing remembered task content reduces success even when account and object identifiers remain available. Replacing that content redirects actions toward the altered record, showing how the information retained by the agent shapes the transaction it executes.

The experiments distinguish having information, using it and acting correctly. If a saved record supplies information needed for an action, removing it should impair execution and restoring it should recover performance. Removing unrelated text tests whether the effect reflects general disruption. Retaining identifiers while removing other content separates tool-address information from task facts. Replacing a task fact tests whether the action follows the changed value; scoring against both the original and changed target identifies this directional response even when it produces an incorrect action. Making the same fact available elsewhere tests whether another source can compensate for memory loss. These comparisons extend the intervention logic used to study reasoning traces and representations to records retained by an acting agent (Turpin et al., 2023; Lanham et al., 2023; Geiger et al., 2021).

The study proceeds from controlled action choices to transactions executed through tools. In the latter experiments, the agent writes its own state, and we save a checkpoint immediately before execution. Each intervention starts from that same checkpoint. Figures 3–5 present the resulting evidence for state dependence, recovery and content use. A software-localization application asks a further question: can responses to state changes help predict whether an agent has identified the correct files, beyond information from additional generic queries? This connects the behavioral account of state use to the practical assessment of an agent’s answers.

2. Related work

Task benchmarks evaluate agents against externally specified objectives. AgentBench, WebArena, τ -bench and SWE-bench cover interactive decisions, policy-governed tool use and software repair (Liu et al., 2024; Zhou et al., 2024; Yao et al., 2025; Jimenez et al., 2024). ToolSandbox evaluates intermediate and final milestones in environments with stateful tool dependencies (Lu et al., 2025). These settings make execution outcomes observable, including failures that plausible verbal answers can conceal. Our experiments use such outcomes to compare actions from a common checkpoint under different versions of a saved record.

Memory evaluation further distinguishes retaining information from applying it. Mem2ActBench constructs tasks whose tool parameters depend on facts distributed across prior interactions, directly testing memory-supported parameter grounding (Shen et al.,

2026). Analyses that vary writing and retrieval strategies locate different bottlenecks in access to stored information and its utilization (Yuan et al., 2026). The present comparison fixes the record actually written by an agent and varies its availability, content and source of access. This permits task facts to be separated from transaction identifiers and tests whether earlier history or redundant facts can substitute for the removed record.

Intervention studies of reasoning supply the closest methodological basis. Truncation, inserted mistakes and paraphrases test different forms of dependence on a chain of thought (Lanham et al., 2023); mediation analyses distinguish the effects of a reasoning problem and its intermediate reasoning on an answer (Paul et al., 2024). Interchange interventions test whether replacing a representation produces the behavior associated with the substituted value (Geiger et al., 2021). This yields a specific behavioral expectation for saved state: substituting a task-relevant value should move the action toward the corresponding alternative target. We test this expectation in exposed memory and goals. Scoring each action against both the original and altered task target distinguishes movement toward the replacement from an undirected decline in success. The intervention concerns the record supplied to execution, with the continuation policy held fixed.

Controlled replay provides a complementary account of agent behavior. Causal Agent Replay estimates shifts in outcome distributions after interventions at selected steps and treats policy sampling as part of the comparison (Shah, 2026). Process-evaluation and executed-replay analyses likewise separate action prediction, task uncertainty and step contribution (He et al., 2026; Zhang, 2026). Our sampling unit is a task with a saved checkpoint: each condition receives fresh continuations from the same environment. Relevant deletion, irrelevant edits, restoration and content substitution characterize the distribution of executable outcomes conditional on the saved state and continuation policy.

Reliability estimation asks what generated responses reveal about correctness. Self-consistency aggregates repeated answers (Wang et al., 2023); semantic uncertainty groups answers by meaning before measuring their variation (Kuhn et al., 2023; Farquhar et al., 2024); SelfCheckGPT uses agreement across sampled generations as evidence about factuality (Manakul et al., 2023). These methods motivate a demanding comparator: extra generic observations can themselves be informative. The localization analysis therefore gives intervention and generic diagnostics matched information, learner-selection opportunities and recorded token funding. Development and independent evaluation are separated to distinguish representation selection from generalization (Cawley and Talbot, 2010). File localization and patch resolution remain distinct outcomes, following the separation of these tasks in software-repair systems (Xia et al., 2024).

3. Definition and study design

3.1. State dependence and intervention contrasts

A saved record can influence an action in several ways: it can supply a needed fact, identify the object on which a tool operates, or specify which feasible action serves the goal. We hold the agent’s completed state write and the execution environment fixed, then vary what the continuing agent can read (Figure 1). The unaltered record provides the reference condition, termed sham; a targeted deletion is termed a lesion. Comparing fresh

continuations from the same checkpoint isolates the effect of that change under the chosen continuation policy.

For task e , let h_e denote the context supplied at the execution checkpoint, x_e the resettable environment, and $s_e = (m_e, g_e, u_e)$ the agent-written memory, persistent goal and unrelated record. Intervention I_b changes the designated state component before a fresh continuation with model randomness ϵ . Its scored outcome is

$$Y_{eb} = W_e f\{\text{Run}_\pi(h_e, I_b(s_e), x_e; \epsilon)\}, \quad (1)$$

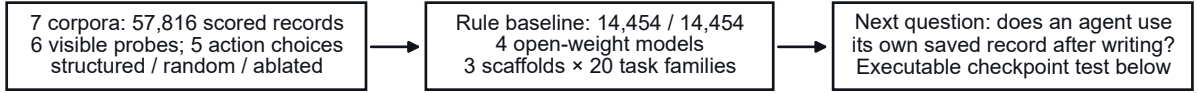
where f is an executable evaluator hidden from the model and W_e is the indicator of a valid state write. Raw environment success is retained separately from this preparation-gated primary endpoint. For component k , the primary paired effect is

$$\Delta_k = \mathbb{E}_e [\Pr(Y_{e,\text{sham}} = 1) - \Pr(Y_{e,\text{lesion}(k)} = 1)]. \quad (2)$$

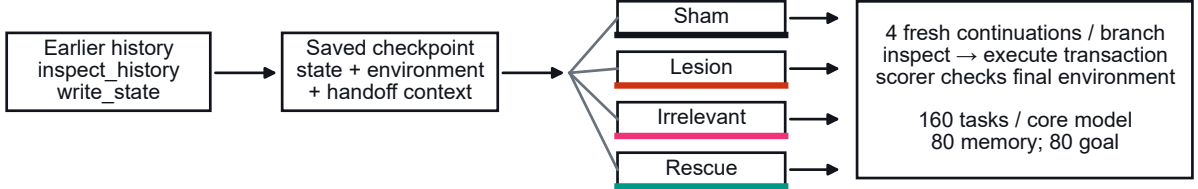
The expectation gives equal weight to the eight task mechanisms and then to their pre-specified task instances. A positive Δ_k means that removing component k reduces execution success. Restoration tests whether the agent’s own original record recovers that success, while deletion of unrelated text tests sensitivity to context disruption. These comparisons are interpreted alongside the quality of the written state and the agent’s unperturbed task performance. Four fresh continuations per condition estimate each task’s success probability.

Alternative access and content replacement help explain a deletion effect. If restating a fact at handoff or reopening the earlier history compensates for memory loss, the dependence reflects access to that information. If replacing a value redirects the action toward its associated target, the record controls what the agent does as well as whether it succeeds. The finite-action scores measure responses to supplied rules; the action-field coupling index (AFCI) measures agreement between reported fields and actions with knowledge of the fields’ source. CSB denotes the state–action relation, with these scores and the execution contrasts providing distinct evidence about it (Table 1).

a Controlled information: explicit rules and matched prompt controls



b Execution: intervene after the agent has written its state



8 mechanisms x 20 tasks; memory / goal = 80 / 80; repeats do not enlarge task n.

c One saved checkpoint, four observed continuations

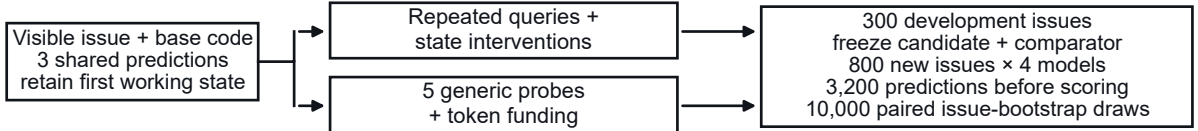
Sol / aggregation / memory 1: AC-407423; REP-9366

Memory: settled entries ≥ 100 cents. Goal: sum by day.

Stored memory	Mon / Tue / Wed (cents)	Original / altered target
Unchanged	750 / 900 / 800	1 / —
Empty dictionary	No accepted transaction	0 / —
Restored	750 / 900 / 800	1 / —
Threshold $\rightarrow 200$	650 / 800 / 800	0 / 1

Same environment reset; independent draws. First nonredundant aggregation-memory checkpoint.

d Prediction: isolate reference-free features from outcome scoring



Reference paths: scorer only, released after all 3,200 predictions are frozen.

Figure 1: From controlled responses to state-dependent action. **a**, Explicit finite-action rules and matched controls establish the controlled response patterns. **b**, Agents inspect earlier history and write their own memory and goal before a checkpoint is saved. Each branch starts from the same environment with fresh model draws; 160 tasks per core configuration comprise 80 memory and 80 goal tasks. **c**, Recorded first continuations for a Sol event-aggregation task, with the relevant facts available only in saved memory. The source data identify the checkpoint and each continuation. The table reports unchanged, removed, restored and substituted memory under the same goal. The substituted record raises the amount threshold from 100 to 200 cents; original and altered targets are scored separately. Dashes mark the branches for which only the original-target score is displayed. This worked example illustrates the intervention semantics, while Figures 3–5 report all tasks. **d**, Predictors are selected on 300 development issues and frozen before 800 new issues are evaluated by four configurations; all 3,200 predictions precede reference-data release.

Table 1: State variables and measured outcomes.

Term	Meaning in this study
Exposed state	The saved record supplied to the continuation at the execution checkpoint.
Memory	Task facts retained from the earlier interaction.
Persistent goal	A saved optimization or conflict-resolution policy that selects among feasible actions.
Execution success	Correct final local environment transaction, including parameters and task policy.
Reference-free diagnostic	Computable from visible inputs and model outputs, without reference answers.
AFCI	A score combining reported content, its source and final-action alignment.

3.2. Agent-written state and tool execution

The execution tasks require an agent to carry information from an earlier interaction into a later tool call. For example, remembered quantities can determine an invoice amount, while a saved conflict-resolution policy can select among feasible appointments. The eight task mechanisms are service allocation, meeting scheduling, directed routing, inventory splitting, invoice arithmetic, dependency scheduling, versioned record merging and grouped event aggregation. After writing its state, the agent inspects the operational environment and submits a transaction with task-specific parameters. An executable evaluator checks the resulting stock, intervals, paths, amounts, dependencies or records. All tasks use the same compact JSON tool interface.

Sol, DeepSeek, Qwen2.5-14B-Instruct and Mistral-7B-Instruct-v0.3 each receive 160 tasks, divided equally between memory and persistent-goal interventions. The two open-weight models use quantized weights. Astra is evaluated on a matched 40-task subset. Every core condition receives four fresh continuations from the saved checkpoint. Thirty-two tasks restate the relevant information at handoff; a fixed 40-task subset also tests paraphrase, replacement of one field and access to earlier history. Sham and restoration supply the same agent-written values, with independent model draws. The content experiment then separates remembered facts from transaction identifiers on 128 additional tasks per API configuration. Methods 5.2 gives the model specifications, preparation requirements and inference settings.

3.3. Reference-free issue-localization prediction

The software application asks whether an agent’s response to a changed working hypothesis helps predict the correctness of its file localization. Each issue is presented with a repository at a fixed commit and source excerpts retrieved from that version. Three initial calls each propose up to three implementation files and a working hypothesis. Localization succeeds when at least two of those calls each identify a reference implementation file. The first call supplies the saved hypothesis used by the diagnostics, without selection by confidence or correctness. Reference patch paths, test results and outcome labels are available only to the evaluator.

We compare state interventions with additional generic queries about the same issue. Both strategies receive the same issue, source excerpts and original working hypothesis. The repetition–intervention diagnostic combines three reevaluations of the current state, one response after state removal and one after a mismatched state. This combination measures ordinary response variation alongside changes elicited by an intervention. The generic comparator starts with five queries and can receive additional queries to match the diagnostic token cost. Feature-representation choices, learner families and tuning opportunities are the same for both strategies. A simpler linear comparison tests seven features per strategy; Methods 5.3 defines both representations and their budget rules.

Predictors are developed on 300 SWE-bench Lite issues and then fixed before evaluation on 800 unused issues from the full SWE-bench test population, stratified across 12 repositories. Previously used issues and near-duplicates are excluded. Each new issue is evaluated by all four core models, yielding 3,200 predictions fixed before reference information is released for scoring. This design tests whether a diagnostic learned from development tasks predicts success on new issues. A separate 100-issue reassessment and the reference-dependent analysis have different evaluation roles, specified in Methods and Supplementary Note 1.

4. Results

4.1. *Structured information supports controlled action responses*

Structured information produced a consistent action-response advantage in the finite-action experiment (Figure 2). Across 57,816 scored records from seven corpora, the structured condition exceeded the high-randomness comparator in every corpus under the registered composite criterion. The mean contrast was 0.886, with a descriptive two-sided 95% Student t interval of [0.843, 0.929] across the seven corpus differences. Reason and veto removal also changed their corresponding responses in the registered directions. The corpus is the inference unit for these comparisons; shared templates and scoring rules define their common experimental setting.

The four-model control experiment asks which supplied information drives this advantage. Across Qwen2.5-7B, 14B and 32B and Mistral-7B, structured inputs outperformed context-field removal in 238/240 combinations of prompt scaffold and task family, scrambling in 240/240 and the distribution-matched control in 236/240. Their mean contrasts were 0.550, 0.895 and 0.593, respectively. Ordinary state-field deletion produced a smaller mean contrast of 0.041, positive in 193/240 combinations. In the additional Qwen-32B experiment, strict removal of the decisive field produced a contrast of 0.961, positive in all 60 combinations. The effect therefore depends on which information is removed. These summaries describe variation within the shared experimental design.

The structured-response advantage also persisted in a local comparison with exact prompt padding and fixed generation length: Qwen2.5-1.5B accuracy was 0.900 versus 0.325. Thus, this comparison holds both prompt and generation budgets constant. The broader entropy calibration addressed a different question, whether output distributions could be made equally variable: its smallest maximum gap was 1.171 bits, above the prespecified 0.15-bit threshold. Prompt-token alignment alone did not equalize total tokens or latency. Supplementary Figures S7–S8 report the budget-controlled responses and the entropy-calibration profiles.

The visible rules provide a direct account of performance in this setting. A deterministic baseline, given the same rule template and module flags, reproduced all 14,454 structured-condition targets, with accuracy 1.000. The scorer’s target variable was withheld from generation, but the correct response was recoverable from the visible rule. Thus, the finite-action experiment establishes dependence on explicitly supplied information. The tool-execution experiment addresses the corresponding question for records produced by the acting agent itself.

The five available responses were action A, action B, veto, deferral and recall of a prior commitment. Malformed or unmatched responses were recorded as invalid. Table 2 gives the probe-to-action mapping; deferral is permitted by the interface but is not a target of these six probes. The synthetic open-weight control matrix uses a separate four-action set. The canonicalization procedure and response-score definitions are given in Methods.

Table 2: Visible prompt-to-action rules in the finite-action experiment. The module flag and rule are available to both the model and the deterministic baseline.

Probe	Relevant module enabled	Module disabled
Baseline / irrelevant cue	<code>ACTION_A</code>	<code>ACTION_A</code>
Reason flip	<code>ACTION_B</code>	<code>ACTION_A</code>
Memory conflict	<code>RECALL_PRIOR</code>	<code>ACTION_A</code>
Veto cue	<code>VETO</code>	<code>ACTION_A</code>
Self-continuity	<code>ACTION_A</code>	<code>ACTION_B</code>

A decisive-field positive control further tested the information responsible for the choice. It comprised 1,440 API generations across 20 task families, six events per family, six conditions and two models. Both full-state and decisive-field-only accuracy were 1.000 for GPT-5.4-mini and Qwen3-max; the strongest non-decisive control, action-prior-only, reached 0.208. A local consistency guard checked the event identity of the decisive field and its agreement with the action, correcting locally or deferring when the field belonged to another event. Without additional model calls, following of scrambled fields fell from 0.958 to 0.033 in GPT-5.4-mini and from 0.533 to 0.033 in Qwen3-max, while irrelevant-cue accuracy remained 1.000. The guard demonstrates a concrete use of event-specific information where the action rule is explicitly available.

The corpus distributions, replicate values and leave-one-corpus summaries are reported in Supplementary Figures S4–S5; Figure S9 gives the complete open-weight comparison. The individual corpus and replicate values show the variation underlying the aggregate contrasts.

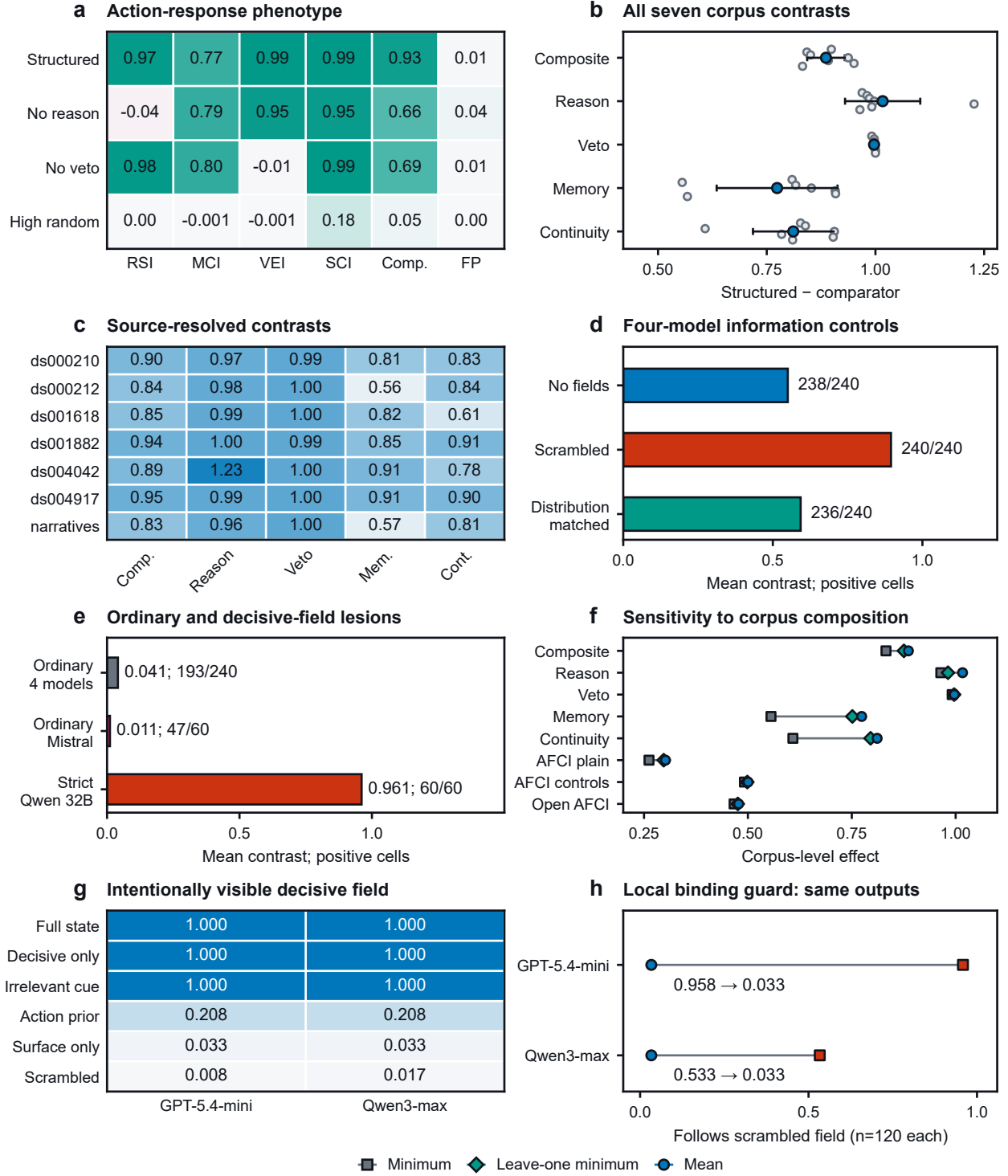


Figure 2. Controlled information and action responses.

Panel b: seven-corpus response-score contrasts with descriptive 95% Student t intervals ($df = 6$).

Figure 2. Detailed legend.

Controlled information and action responses. **a**, Behavior-level reason (RSI), memory (MCI), veto (VEI), self-continuity (SCI), composite and irrelevant-cue false-positive (FP) scores. **b,c**, The same five response-score contrasts across seven corpora, summarized in b and resolved by corpus in c. Reason and veto compare structured with the corresponding component deletion; composite, memory and continuity compare structured with high-random output. Differences have range $[-2, 2]$. Descriptive two-sided 95% Student t intervals (6 degrees of freedom) and source heterogeneity; the original percentile-bootstrap criterion is reported separately in SI Table S45. **d,e**, Open-weight mean contrasts and positive scaffold-task-family cells; strict deletion is a separate Qwen-32B slice. **f**, Minimum, leave-one-corpus minimum and mean effects; colors and shapes follow the bottom key. **g**, Decisive-field positive-control accuracy (120 events/condition/model). **h**, Scrambled-field following before (red square) and after (blue circle) the local guard, on the same 120 outputs/model without an added call. Cell counts are descriptive; corpora share templates. Full profiles remain in SI Figures S4, S5 and S9.

4.2. Agent-written memory and goals support recoverable execution

The execution experiment tests whether the same dependence occurs when the agent has written the record itself. Removing memory or a persistent goal reduced transaction success in Sol and DeepSeek (Table 3; Figure 3). Across the 80 tasks per component, Sol’s memory and goal effects were 0.750 and 0.366; DeepSeek’s were 0.541 and 0.259. Each task contributes four continuations per condition. All four contrasts passed the prespecified eight-comparison Holm adjustment ($p < 0.001$), and their unperturbed success rates ranged from 0.675 to 0.938.

The quantized open-weight configurations had lower baseline competence. Qwen’s effects were 0.081 for memory and 0.044 for goal (adjusted $p = 0.031$ and 0.137), with unperturbed success rates of 0.106 and 0.125. Mistral completed only 1/320 unperturbed memory continuations and 4/320 goal continuations. Its small contrasts therefore occur at floor performance, where changes in an already rare success event have limited diagnostic resolution.

Table 3: Execution after agent-written state. Effects are sham minus targeted-lesion success with equal mechanism weights. Intervals resample tasks within mechanisms; sham is the unweighted observed success rate. All planned preparation and execution failures are included.

Model	Component	Tasks	Sham	Effect	Two-sided 95% CI
Sol	memory	80	0.925	0.750	[0.656, 0.834]
Sol	goal	80	0.938	0.366	[0.269, 0.462]
DeepSeek route	memory	80	0.675	0.541	[0.453, 0.622]
DeepSeek route	goal	80	0.681	0.259	[0.184, 0.338]
Qwen 14B	memory	80	0.106	0.081	[0.034, 0.134]
Qwen 14B	goal	80	0.125	0.044	[0.009, 0.084]
Mistral 7B	memory	80	0.003	0.003	[0.000, 0.009]
Mistral 7B	goal	80	0.013	0.006	[0.000, 0.019]

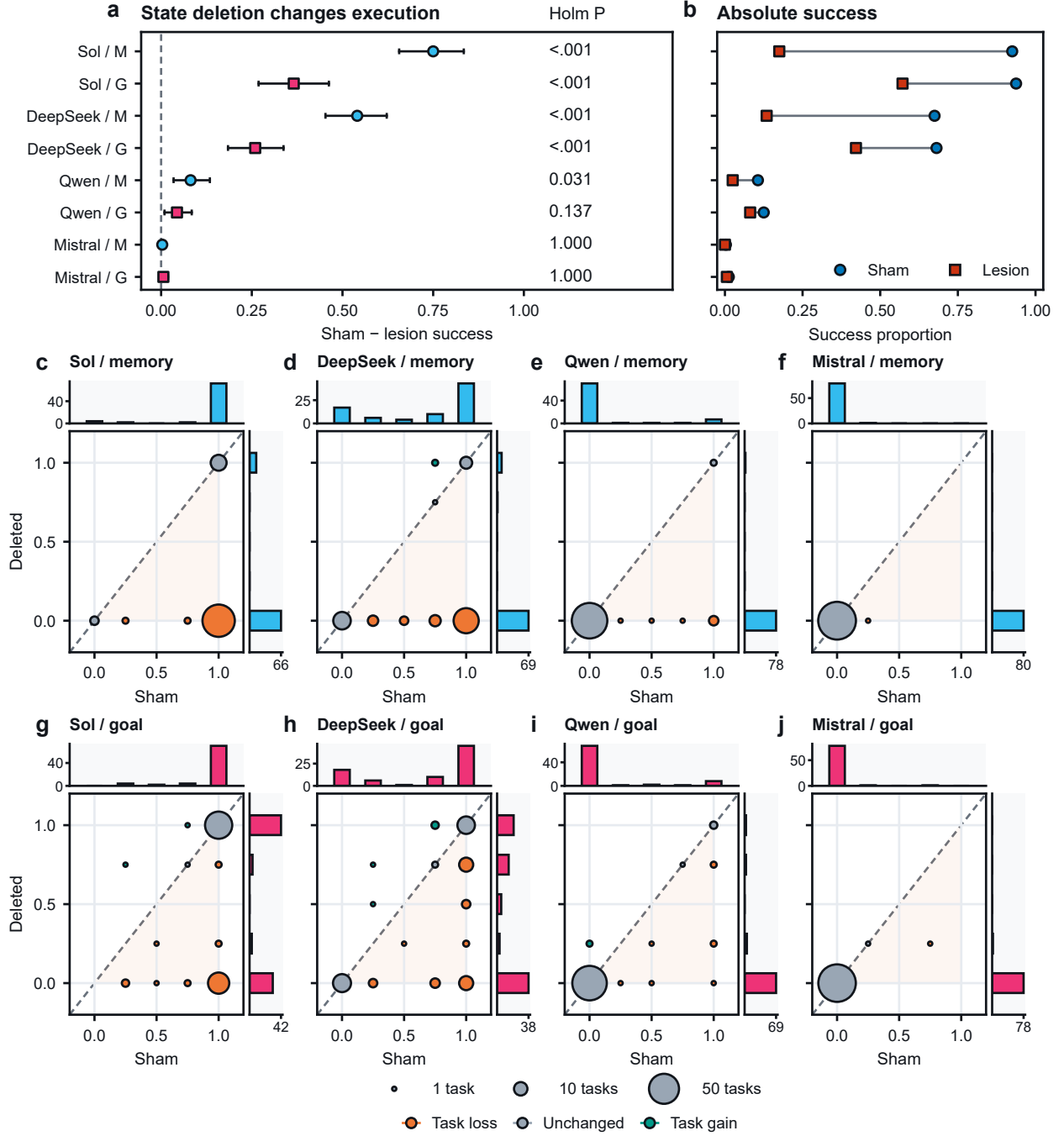


Figure 3: State dependence, absolute competence and paired task outcomes. M and G denote memory and persistent goal. **a**, Registered equal-mechanism sham-minus-lesion effects and unadjusted two-sided 95% task-stratified intervals (80 tasks per component/model). Displayed p values are Holm-adjusted across the eight primary contrasts. **b**, Absolute sham and lesion success over 320 continuations each. **c–j**, All 80 paired task means per panel, averaging four continuations per branch; the first row is memory and the second persistent goal. Bubble area is proportional to task count at each attainable pair of success fractions. Orange marks lower lesion success, teal higher lesion success and gray ties; the dashed diagonal denotes equality. Aligned top and right histograms show the respective marginal task counts. Cyan and magenta identify memory and persistent goals in a and in the marginal histograms. The baseline success rates in b provide the context for the deletion effects. All planned cases, including failures, remain in the denominator. The task is the paired analysis unit. SI Figures S1–S3 resolve mechanisms, matched Astra subsets and failure intersections.

Restoration and unrelated edits distinguish dependence on the saved information from disruption caused by editing context. Restoring the agent’s original record recovered most of the success lost under deletion: the memory and goal recovery effects were 0.753/0.366 in Sol and 0.519/0.238 in DeepSeek. Unrelated edits changed their success rates little, with sham-minus-irrelevant differences from -0.013 to 0.006 . Directly comparing unrelated edits with targeted deletion gave memory effects of 0.744 [0.656, 0.828] for Sol and 0.538 [0.456, 0.619] for DeepSeek; the goal effects were 0.359 [0.269, 0.453] and 0.272 [0.191, 0.353] (Figure 4). These four supplementary comparisons passed their separate Holm adjustment ($p < 0.001$). Qwen showed a smaller memory effect, 0.078 [0.031, 0.128] ($p = 0.015$), with $p = 0.090$ for goal; neither Mistral comparison established a difference.

Dependence also varied with alternative access to the information. Providing earlier history increased memory-lesion success by 0.563 in Sol and 0.438 in DeepSeek. When relevant facts were already restated at the execution checkpoint, the corresponding memory-lesion effects were 0.000 and -0.016 . Together with restoration, these controls show that access to task-relevant information explains more of the observed pattern than the mere presence of a particular storage field. Paired descriptive intervals summarize the control residuals; encoding equivalence is evaluated separately in the content experiment.

Replacing saved content redirected transactions toward the corresponding alternative. We scored each original-state and replacement-state action against both the original and alternative task target. For memory, Sol actions matched the original target in 75/80 original-state continuations and 19/80 replacement continuations, while matches to the alternative target rose from 4/80 to 61/80. DeepSeek showed the same directional pattern: original-target matches fell from 57/80 to 7/80, and alternative-target matches rose from 3/80 to 56/80. The symmetric target-alignment contrasts were 0.706 and 0.644 for memory and 0.719 and 0.638 for persistent goals, respectively. These crossed comparisons identify the direction of state-dependent action alongside the deletion and restoration effects. Qwen showed little target-specific redirection and Mistral remained at floor performance. The supplementary analysis retains overlapping and undefined alternative targets and distinguishes redundant from nonredundant access to the original information.

Across these controls, the clearest pattern is loss after targeted deletion, recovery after restoration or renewed access, and action redirection after content replacement. Figures 3 and 4 show the paired tasks and controls; Supplementary Figures S1–S3 resolve the eight mechanisms, the matched 40-task Astra comparison and preparation and execution failures.

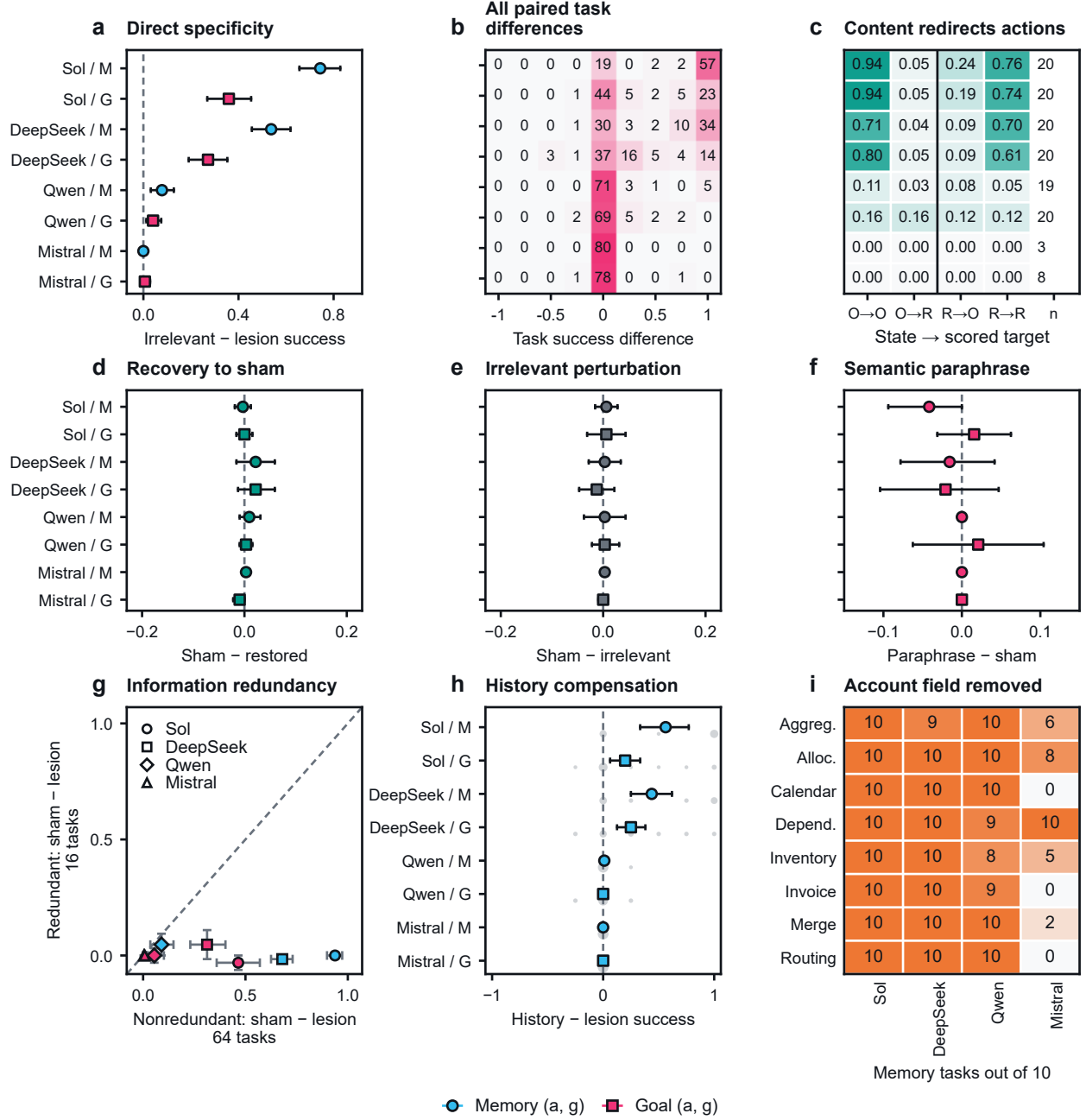


Figure 4: Specificity, recovery and the direction of state-dependent action. Memory (M) and goal (G) row order is shared in a–f and h. **a**, Irrelevant-minus-lesion effects with 95% intervals from 5,000 task-stratified supplementary bootstrap draws (80 tasks/cell). **b**, Exact task counts at the nine attainable paired differences; every row contains 80 tasks. **c**, Actions generated under original (O) or replaced (R) state, scored against each target: O→O, O→R, R→O and R→R correspond to $S_{00}, S_{01}, S_{10}, S_{11}$. The column n gives tasks with a defined alternative target, out of 20 planned; each contributes four continuations per state. **d,e**, Sham-minus-restored and sham-minus-irrelevant residuals (80 tasks). **f,h**, Paraphrase-minus-sham and history-minus-lesion contrasts on 20-task subsets; pale point areas in h show frequencies of individual task differences. Circles denote memory and squares persistent goals in these contrast panels. **g**, Nonredundant versus redundant effects with separate 95% intervals (64 and 16 tasks); colors distinguish components, shapes configurations, and the diagonal indicates equality. **i**, Memory records whose account field is removed by the original lesion, out of ten tasks per mechanism/configuration (266/320 overall). All displayed intervals are two-sided 95%. Figure 5 isolates memory content while preserving identifiers; its panel g compares encoding controls.

4.3. Memory content contributes beyond transaction identifiers

Deleting a whole memory also removes identifiers needed to address a tool call. To separate that effect from loss of task facts, we retained account, object and other identity fields while removing the remaining content. Sol and DeepSeek were evaluated on 128 tasks across the eight mechanisms, with eight memory conditions and two continuations per condition; task identities and analysis were fixed before acquisition. Success fell from 206/256 to 38/256 in Sol and from 169/256 to 24/256 in DeepSeek. The paired probability differences were 0.656 (95% task-bootstrap interval [0.594, 0.715]) and 0.566 [0.527, 0.605], respectively; both two-model Holm-adjusted sign-flip tests gave $P < 0.001$ (Figure 5). Task content therefore contributed to execution even when transaction identifiers remained available.

The content-loss effect was positive in all eight mechanism-specific point estimates, with magnitudes varying across mechanisms and models. Identity-only memory lowered the two-continuation success fraction on 103 of 128 Sol tasks and 86 of 128 DeepSeek tasks; 22 and 40 were unchanged, and 3 and 2 increased. Identifiers were also necessary for successful execution in this sample: neither content-only nor empty memory produced a successful transaction. Restoring the original record yielded 215/256 successes in Sol and 167/256 in DeepSeek, while irrelevant editing yielded 215/256 and 166/256. Both controls remained close to full-memory performance relative to the large content-removal effects. All 4,096 planned continuations were included, including those following one unsuccessful DeepSeek state write. The content-loss contrast remained positive under the extreme assignments of terminal service failures reported in Supplementary Methods.

Encoding controls separated the availability of facts from their presentation. Exact JSON text preserved the complete saved memory while changing its container from a structured object to a string. In Sol, it increased success by 0.074 [0.020, 0.129] relative to the structured record. Deterministic rewording preserved the same values; its difference from exact JSON text was -0.094 $[-0.148, -0.039]$. In DeepSeek, the reworded-minus-JSON contrast was 0.008, with a 90% interval of $[-0.027, 0.042]$ inside the prespecified descriptive equivalence range of ± 0.05 . Rewording differed little from the original structured record in the overall point estimates (-0.020 in Sol and -0.012 in DeepSeek). The experiments therefore distinguish a large dependence on remembered task content from smaller, model-dependent effects of its encoding.

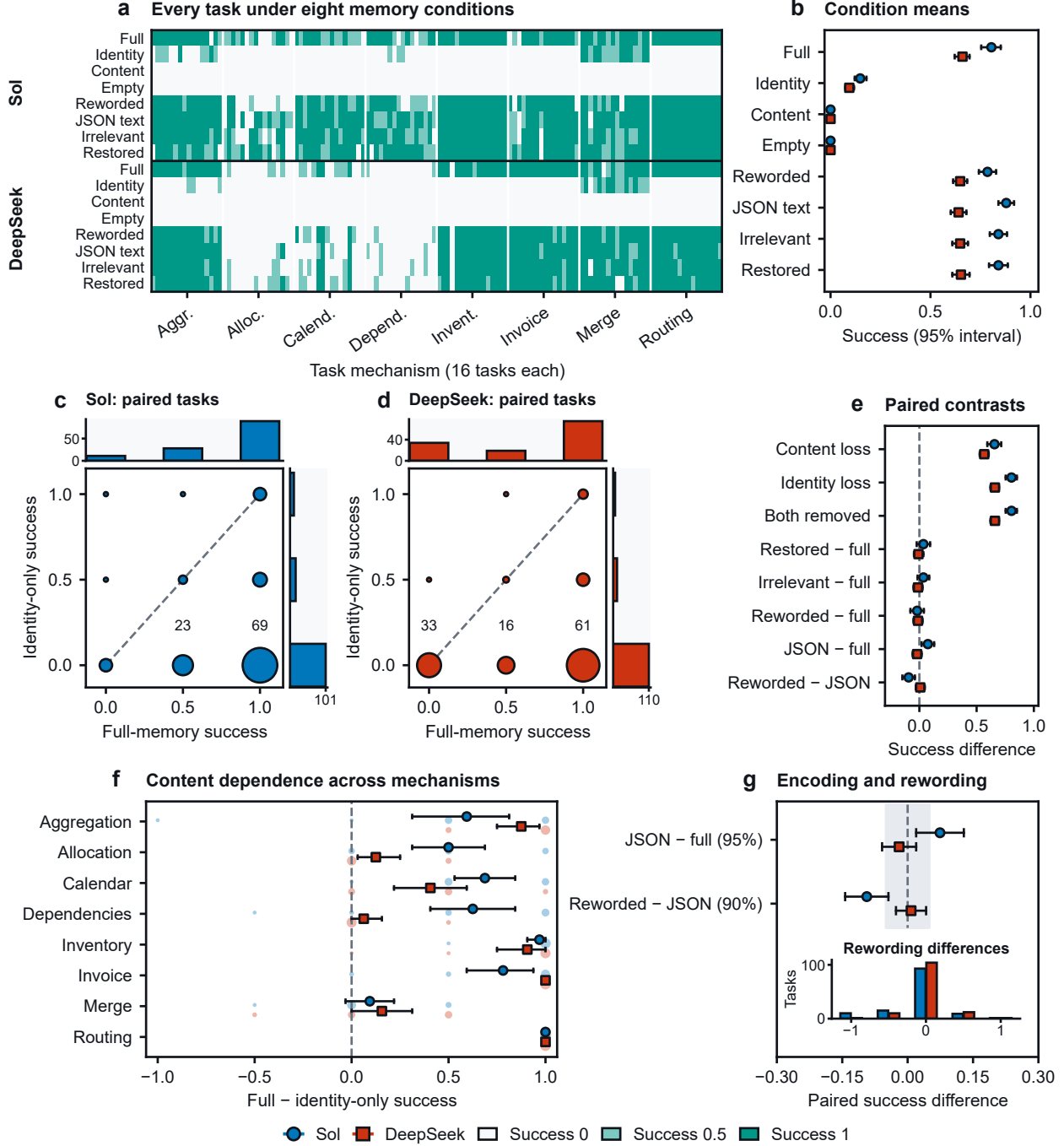


Figure 5: Memory content supports execution when identifiers remain available. **a**, All eight conditions for the same 128 new tasks, ordered within eight mechanisms. Separate Sol and DeepSeek blocks show two-continuation task means; colors encode success fractions 0, 0.5 and 1. **b**, Condition means with 95% task-bootstrap intervals (256 continuations/condition/model). **c,d**, Full-memory versus identity-only task means. Bubble area denotes task count on a common scale; cells with at least 16 tasks are labeled. Dashed lines indicate equality; marginal histograms retain all 128 tasks. **e**, Paired contrasts with 95% intervals. Content loss, identity loss and both removed subtract identity-only, content-only and empty-memory success from full-memory success, respectively. Intervals in **b** and **e** use 10,000 mechanism-stratified draws; the two primary content-loss tests use 50,000 sign flips with Holm adjustment across models. **f**, Mechanism-specific content-loss effects with descriptive 95% intervals over 16 tasks. Pale point areas show frequencies of attainable task differences. **g**, JSON-text-minus-full contrasts use 95% intervals; reworded-minus-JSON contrasts use 90% intervals for the registered descriptive equivalence comparison. The gray band marks ± 0.05 ; the inset shows all discrete task-level rewording differences. Denominators include failed preparations and terminal service failures; zero-width intervals describe floor observations.

4.4. State diagnostics and localization reliability

The localization study tests whether responses to state interventions help predict correctness beyond additional generic queries. On 800 previously unused issues from 12 repositories, the repetition–intervention diagnostic achieved AUC 0.9403, compared with 0.9365 for the token-funded generic comparator (Table 4). All 3,200 planned records from Sol, DeepSeek, Qwen and Mistral entered the evaluation. The paired difference was 0.00382, with a marginal 95% issue-bootstrap interval $[0.00023, 0.00743]$. The prespecified three-comparison 98.333% interval was $[-0.00055, 0.00821]$, so the pooled increment did not meet the family-level superiority criterion. Figure 6 shows this comparison and its variation across configurations.

The development comparisons identify the response information used by the predictors. On the 300-issue cohort, retaining richer response relationships and averaging four learner families gave AUC 0.9480 for repetition–intervention and 0.9433 for the equally enhanced generic comparator. With seven-feature linear representations, the corresponding AUCs were 0.9235 for state probes and 0.9245 for token-funded generic probes, with paired full-refit difference -0.00097 $[-0.01317, 0.00473]$. In a matched ablation, output-quality features gave AUC 0.9433, whereas quality plus relations between answers gave 0.9487; adding state correspondence alone did not recover that gain. Each strategy had the same representation and tuning choices. These development results informed the predictors fixed before independent evaluation. Supplementary Methods report the complete ablations, and Figures S12–S14 give the linear comparison and 100-issue reassessment.

Table 4: Independent localization prediction on 800 new issues and four configurations (3,200 model–issue records). All predictors were frozen before the new outcomes were scored. Higher AUC and lower Brier score or log loss indicate better prediction. The primary paired AUC difference is 0.00382 (95% interval [0.00023, 0.00743]; prespecified family 98.333% interval [−0.00055, 0.00821]). Paired uncertainty and the comparison family are shown in Figure 6.

Diagnostic	AUC	Brier	Log loss
Five generic queries	0.9364	0.0927	0.2929
Token-funded generic	0.9365	0.0929	0.2936
State interventions	0.9382	0.0911	0.2901
Repetition–intervention	0.9403	0.0900	0.2873

The predicted probabilities also had lower average squared error. Brier score was 0.09000 for repetition–intervention and 0.09293 for funded generic diagnostics, a paired difference of −0.00293 with marginal 95% interval [−0.00566, −0.00020]. The log-loss difference was −0.00624 [−0.01324, 0.00076]. These prespecified secondary measures assess probability accuracy, complementing AUC’s assessment of ranking. Figure 7 relates predictions to observed success, shows how many observations support each calibration bin, and evaluates failure rates when retaining answers with higher predicted reliability.

Predictive gains varied across configurations. The primary AUC difference was 0.03699 [0.01318, 0.06103] for Qwen, 0.00430 [−0.00401, 0.01287] for DeepSeek, −0.00721 [−0.02631, 0.01151] for Sol and −0.00071 [−0.00496, 0.00327] for Mistral. The subgroup intervals are descriptive marginal summaries. Sol contributed 698 positive outcomes, DeepSeek 564, Qwen 531 and Mistral 20, each out of 800. The pooled AUC combines within-model and between-model differences; Mistral’s model-specific estimate is sensitive to its small number of successes.

The pure state-intervention ensemble and the noise-contrast variant gave AUC differences of 0.00168 and 0.00194 against their corresponding funded comparators. Both marginal and family-adjusted intervals crossed zero. The repetition–intervention combination had the largest pooled point increment.

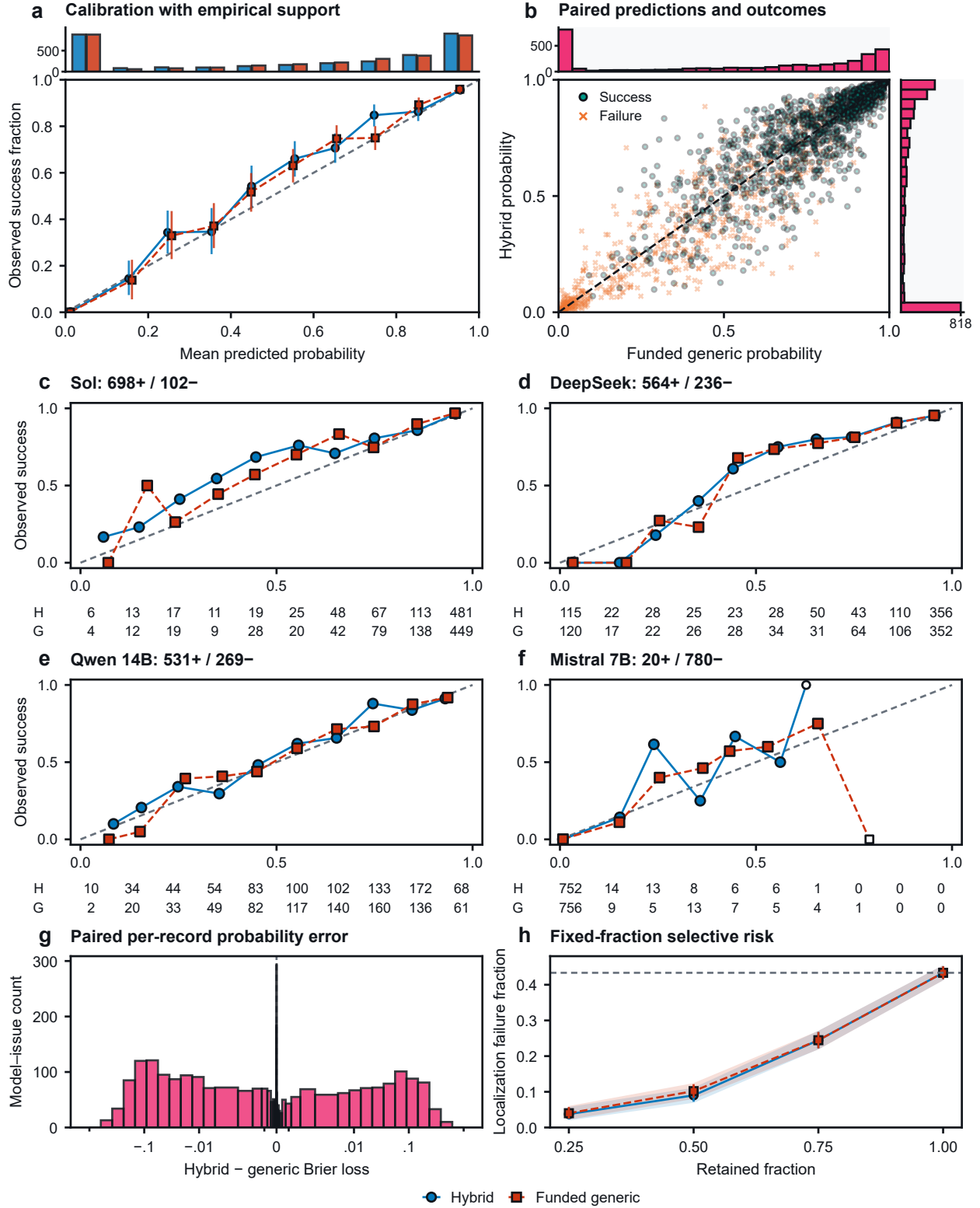


Figure 7. Calibration, paired probabilities and selective error on the independent cohort. 3,200 records; ten fixed bins; pooled and risk intervals: 95% paired issue bootstrap.

Figure 7. Detailed legend.

Calibration, paired probabilities and selective error on the independent cohort. **a**, Observed success versus mean predicted probability in ten fixed bins, with issue-bootstrap 95% intervals and an ideal-calibration diagonal. The aligned histograms give the actual bin counts. **b**, All 3,200 paired predictions, marked by teal circles for success and orange crosses for failure, with empirical marginal probability distributions; the diagonal denotes identical predictions. **c–f**, Model-resolved calibration; rows H (hybrid) and G (funded generic) give all ten fixed-bin counts, including zeros. Empty bins have no point; hollow markers identify bins with one record. These subgroup points are descriptive. **g**, All 3,200 paired per-record Brier-loss differences on a symmetric-log scale with linear threshold 0.005 and nonuniform bins; negative values favor repetition–intervention. **h**, Prespecified retained fractions of 25, 50, 75 and 100%, with failure rates and 95% issue-bootstrap intervals. Ties use the frozen deterministic rule, and risk intervals condition on the observed score selection. The new outcomes are used for evaluation throughout. Blue circles denote repetition–intervention and red squares funded generic diagnostics in a, c–f and h; panel b uses a separate outcome key.

The comparison provided the generic strategy with enough queries to match the candidate’s diagnostic token use whenever metering and acquisition permitted. Usage was complete and both candidate budgets were reached in 3,121/3,200 records. Across all strategies, 241,102,491 tokens were recorded; 81 attempts lacked usage. Restricting the comparison to complete-and-funded records gave a primary AUC difference of 0.00403 [0.00041, 0.00766]. Comparing with the five-call generic diagnostic on the full cohort gave 0.00392 [0.00044, 0.00739]. The positive pooled point difference therefore also occurred under these two budget comparisons. Figure 8 shows actual token pairs and metering coverage; the full planned cohort remains the primary analysis.

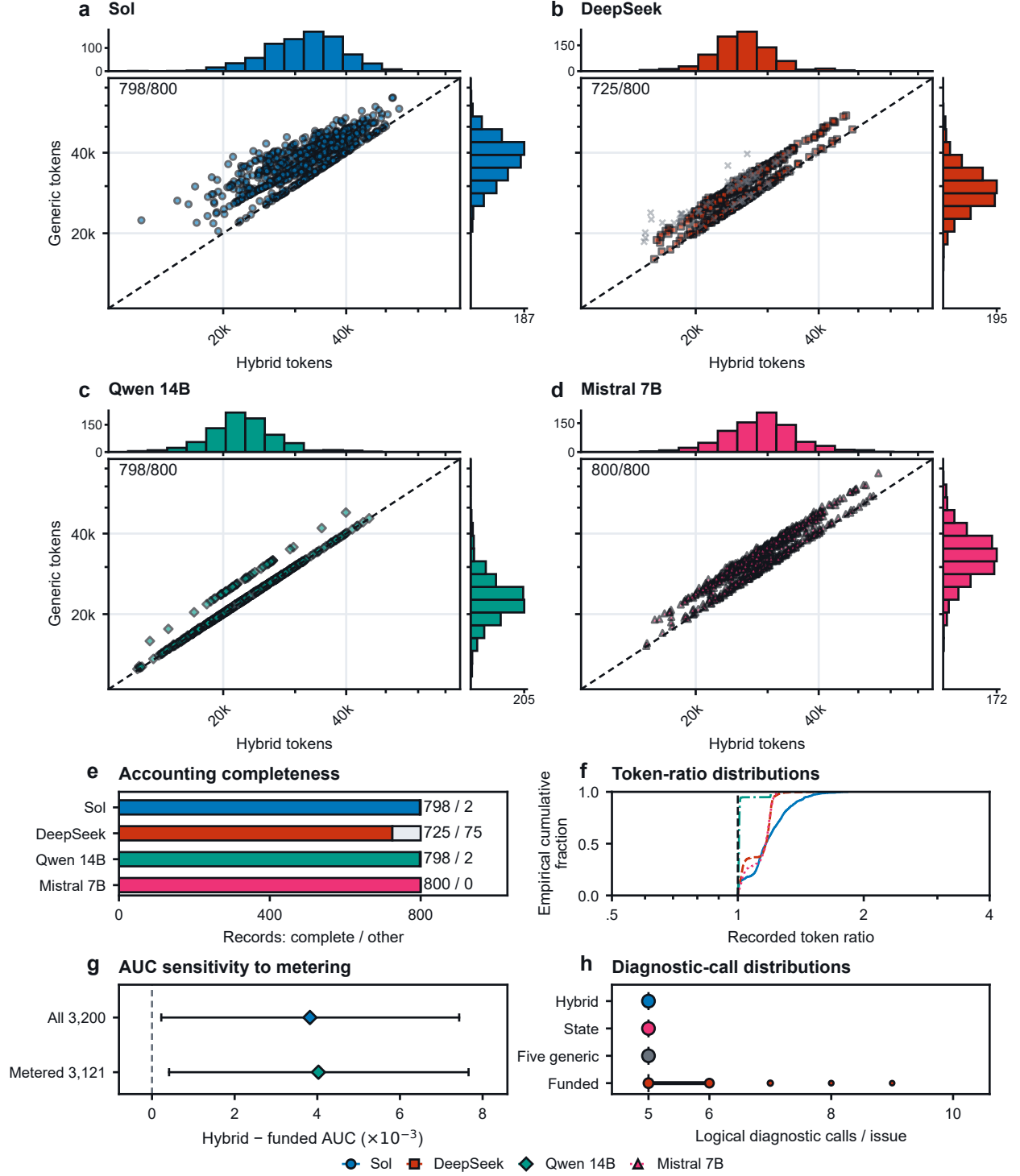


Figure 8: Resource pairing for the independent evaluation. **a–d**, Diagnostic tokens for all 800 model–issue pairs/configuration: repetition–intervention versus funded generic, with common logarithmic axes, equality diagonals and marginal histograms. Gray crosses mark incomplete usage; colored points are complete and funded. **e**, Corresponding counts by configuration. **f**, Empirical generic/candidate token-ratio distributions for positive recorded totals. **g**, AUC differences for all planned records (primary) and fully metered/funded records, with marginal 95% intervals. **h**, Logical diagnostic calls/issue: medians, interquartile ranges and attainable-count frequencies. Shared base calls are charged separately; costs include retries and token overshoot. Eighty-one attempts lack usage, affecting 79 records.

Reference information changes what this prediction problem can establish. When the diagnostic features incorporate reference implementation files, the six-model analysis of 1,800 records gives AUC 0.737 for the ordinary baseline and 0.924 after augmentation, with grouped-validation difference 0.187 [0.127, 0.268]. This larger association describes reference-informed scoring. The independent comparison above instead evaluates diagnostics computable from the issue, visible code and model responses before the reference answer is known.

From localization to repair. The exploratory repair evaluation tests a further practical step beyond locating a relevant file. A relevant file identifies where an edit may be needed; successful repair additionally requires an applicable patch and behavior that passes the repository tests. Of 100 initial model–issue submissions, 44 completed the official harness, one resolved the issue and 56 returned harness errors (Table 5). The local application-and-repair experiment began with 100 nonempty patches and made 93 repair attempts. Eight final patches, spanning seven unique issues, applied locally and completed the harness; none resolved its issue.

Table 5: Official repair evaluation. Counts are model–issue submissions; the filtered applicable subset is reported alongside the full initial denominator.

Stage	Submitted	Completed	Resolved	Harness errors
Initial: GPT-4o-mini	50	27	1	23
Initial: Qwen-plus-latest	50	17	0	33
Initial combined	100	44	1	56
After local apply screen/repair	8	8	0	0

The 92 zero-length final exports include rejected patches. Supplementary Note 4 reports format errors, context or hunk mismatches, missing paths, empty repair responses and other application failures by category. Localization and patch resolution are evaluated as distinct endpoints. A separate software binding-guard condition reduced hard-constraint violations by 0.166 while reducing primary hit@3 by 0.077, documenting a tradeoff between constraint enforcement and retrieval success.

4.5. Output agreement and the information available to a scorer

The execution experiments establish whether changing a record changes an action. The trace analyses ask a complementary question: whether the reported record and action agree in the first place. AFCI averages reason, memory, veto and self-state alignment using both emitted fields and their generation provenance. Its components differ markedly: the structured self-state term is approximately 0.003, while the veto term is near 1. Figure 9 shows these profiles and 35 fixed sensitivity contrasts, comprising equal weights and each leave-one-component-out mean against seven comparators. Reporting components alongside the operational composite shows which kinds of agreement contribute to the average.

When the scorer sees only the emitted text, agreement can be evaluated only where that text supplies enough evidence. A rule-based scorer can assess at least one component in 1.91% of 26,946 traces. An exploratory semantic assessment uses two models to judge 252 outputs sampled across nine variants and seven corpora, producing 504 judgments.

Each judgment relies only on visible content and requires a verifiable quotation. Reason consistency is scorable in 79.4–86.5% of items, memory in 28.2–29.8%, persistent goal in 21.8–28.2%, and veto in 4.4–7.5%. Conditional consistency is generally high among the items that can be scored. Coverage therefore determines how much of the output supports a conclusion about agreement.

The judges also differ in whether they can score an item. With unscorable, inconsistent and consistent treated as three categories, inter-judge κ is 0.410 for reason, 0.810 for memory, 0.759 for persistent goal and 0.403 for veto. Only seven of the 252 outputs are scorable for veto by both judges.

The separate human evaluation gives high agreement on the seven visible-output dimensions. All 25,200 individual ratings were binary. Of 8,400 item–dimension combinations, 8,250 were unanimous and 150 had a 2:1 split, involving 68 items. Across the 21 annotator-pair-by-dimension comparisons, mean Cohen’s κ was 0.961, the minimum individual value was 0.860 and the minimum dimension mean was 0.897. Majority labels use at least two matching votes; agreement uses individual scores before aggregation. The two-annotator, 300-item evaluation uses a distinct sample. Supplementary Figure S10 reports prevalence and disagreement, and Figure S6 gives the complete AFCI profiles.

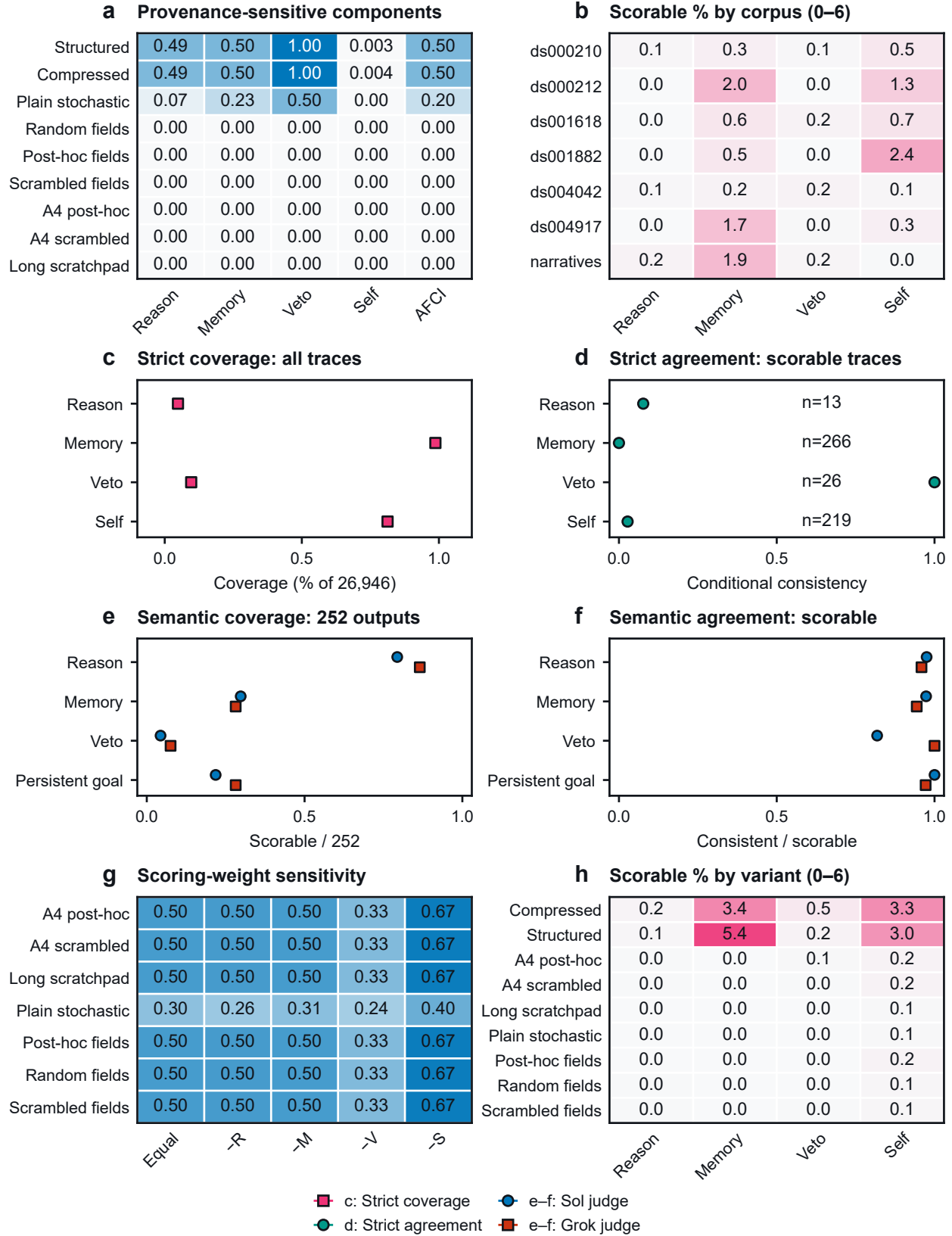


Figure 9. Measurement provenance, coverage and conditional consistency. 26,946 traces; two semantic judges on 252 outputs; descriptive scoring-weight contrasts.

Figure 9. Detailed legend.

Measurement provenance, coverage and conditional consistency. **a**, Content-and-provenance component scores and AFCI, retaining three decimals for sparse self-state scores. **b,h**, Strict output-only scorable percentages by corpus and variant, from all 26,946 traces, with a shared 0–6% color scale. **c,d**, Strict coverage over all traces and consistency among scorable traces, on separate axes with the scorable counts printed in d. **e,f**, Coverage and conditional consistency from the Sol and Grok judges on 252 sampled outputs; the judge field named self evaluates explicit persistent-goal consistency. The denominators differ and these axes are not joined by lines. Legends distinguish strict scoring in c–d from semantic-judge identities in e–f. **g**, Equal-weight and leave-one-component-out AFCI contrasts against seven controls, averaged over corpora; R, M, V and S denote reason, memory, veto and self-state. Weights were not optimized. Full replicate profiles and compression pairs remain in SI Figure S6. AFCI and output-only consistency concern reported fields and their stated provenance.

5. Methods

5.1. Finite-action scores and action canonicalization

The finite-action score asks whether a response preserves the action expected when the relevant component is available. The target is assigned before generation and remains fixed across module variants. For example, the reason-flip target remains action B even when a disabled-reason prompt instructs action A. The score therefore measures preservation of the full-component response; compliance with a variant-specific instruction is a separate quantity. The deterministic baseline in Table 2 was evaluated on all 14,454 structured-condition records.

To score responses consistently, the evaluator maps raw output o_i to a canonical action $\hat{y}_i = g(o_i)$ using the rules in Table 2. It uppercases and trims text, normalizes the specified punctuation and hyphen variants, and matches the five response codes and their aliases in order: ACTION_A, ACTION_B, VETO, DEFER and RECALL_PRIOR. The first match determines the action. Malformed or unmatched responses receive the evaluator code INVALID_OR_UNMAPPED.

Let y_i^- and y_i^+ denote the expected actions before and after the manipulation, and z_i the probe condition. Correct response and irrelevant-cue change are

$$c_i = \mathbf{1}\{\hat{y}_i = y_i^+\}, \quad u_i = \mathbf{1}\{z_i = \text{irrelevant}\}\mathbf{1}\{\hat{y}_i \neq y_i^-\}. \quad (3)$$

For component k and condition v , the operational response score is

$$B_k(v) = \widehat{\text{Pr}}(c_i = 1 \mid z_i \in \mathcal{T}_k, v) - \widehat{\text{Pr}}(u_i = 1 \mid z_i \in \mathcal{N}_k, v), \quad (4)$$

where $\mathcal{T}_k$ and $\mathcal{N}_k$ are the component’s target and irrelevant/placebo probe sets. Each component score lies in $[-1, 1]$, so a between-condition score difference can lie in $[-2, 2]$. Reason, memory, veto and self-continuity scores use their corresponding probe definitions; their composite assigns equal arithmetic weight. The separate trace-based composite and AFCI definitions are given in Supplementary Section S8. The finite-action summaries in Results and Figures 2b and S4b use descriptive Student t intervals over the seven corpus differences: $\bar{d} \pm t_{0.975,6} s_d / \sqrt{7}$, where s_d is their sample standard deviation. The original registered criterion used corpus-cluster percentile bootstrap resampling, retaining within-corpus observations together; its saved 2.5th-percentile endpoint, 0.855732, is reported separately in Supplementary Table S45. The t interval describes the mean corpus contrast, whereas the percentile-bootstrap endpoint evaluates the registered criterion.

5.2. Execution effects and uncertainty

Tasks and model configurations. Task development used 20 initial allocation instances and 20 instances spanning the eight mechanisms; both sets were excluded from the 160-task evaluation. The API request identifiers are `gpt-5.6-sol` and the DeepSeek route `deepseek-v4-pro`; the fixed-weight core uses Qwen2.5-14B-Instruct and Mistral-7B-Instruct-v0.3 using the Q4_K_M weight-quantization format. The provider-served `gpt-6-astra` is allocated a fixed 40-instance subset, shared with the core models. API configurations are identified by their request routes and returned identifiers; open-weight configurations are identified by immutable repository revisions and file hashes. The execution models use a different quantization from the controlled open-weight experiments, so comparisons across those studies also differ in numerical precision.

Open-weight inference used a fixed version of llama.cpp, 15,360 context tokens per slot and Q8-quantized key/value caches. The weight files and evaluation code were identical across the recorded backends. Requests used temperature 0.7 and $\text{top-}p = 1$; open-weight sampling additionally used $\text{top-}k = 40$ and $\text{min-}p = 0.05$, with seeds derived from request identifiers. API continuations used provider sampling, with low reasoning effort for the GPT configurations. Preparation and software queries allowed up to 1,536 output tokens, and execution steps allowed 1,024. Responses were validated as returned against the common JSON interface. The research archive includes complete prompts, runtime defaults, caps and reported usage.

State preparation and execution. Each task begins with history inspection and a state write containing three dictionaries: memory, goal and unrelated material. A deletion empties the selected dictionary while retaining its key. Continuations then inspect the environment and attempt one transaction. Every condition starts from the same saved environment and handoff context, with fresh model draws; condition and repeat order are deterministically shuffled within the checkpoint. Unrelated-content deletion removes material from the agent’s own note, matching the number of serialized characters removed where possible. Character counts and tokenizer-based lengths are reported separately. Supplementary Methods give the exact checkpoint structure and tool-access rules.

The primary endpoint requires both a valid state write and a successful final transaction. Invalid state writes, malformed requests, rejected transactions and missing responses score zero and remain in the planned denominator. Raw environment success is also reported, including correct transactions after an invalid state write. State-write validity and component accuracy are assessed separately; analyses restricted to valid writes are secondary.

The task is the paired analysis unit. Repeated continuations measure variation within that task and do not add independent task instances.

Paired effects and uncertainty. The primary analysis averages paired within-task differences, giving each of the eight mechanisms equal weight and reporting memory and goal separately. It uses 5,000 bootstrap draws resampling tasks within mechanisms and two-sided sign-flip tests with 50,000 draws, a plus-one correction and Holm adjustment across the eight model–component contrasts. The four-model comparison family was fixed before DeepSeek acquisition, after the Sol/Grok/Qwen/Mistral results had been observed. Mechanism-cluster

resampling and leave-one-mechanism-out analyses assess sensitivity to mechanism composition; their resolution is limited by the eight clusters. Supplementary Methods Table M4 gives the shared random stream, statistic and sensitivity specifications.

Restoration, irrelevant deletion, paraphrase, factual replacement and history compensation are secondary controls, summarized by descriptive difference intervals. Their proximity to sham is not treated as an equivalence test. The 128-task content experiment separately evaluates the prespecified rewording-versus-JSON equivalence range of ± 0.05 .

For the supplementary specificity contrast, subtraction is performed within the same task before aggregation:

$$\hat{S}_k = \frac{1}{8} \sum_{f=1}^8 \frac{1}{n_{fk}} \sum_{e \in (f,k)} (\bar{Y}_{e,\text{irrelevant}} - \bar{Y}_{e,\text{lesion}}), \quad \bar{Y}_{eb} = \frac{1}{4} \sum_{r=1}^4 Y_{ebr}. \quad (5)$$

This supplementary contrast was defined after the primary results were inspected. It preserves task pairing and uses 5,000 task-bootstrap draws stratified within task mechanism, seed 2026092601. Two-sided task-sign-flip tests use 50,000 draws and the plus-one correction; Holm adjustment is applied to these eight supplementary contrasts as a separate family. The permutation reference assumes task-difference symmetry under its null; the bootstrap is conditional on the sampled task mechanisms. Replacement continuations are partitioned into original-only, replacement-only, overlapping targets, other valid transactions, invalid transactions and undefined replacement targets. An undefined target takes precedence in this partition and is reported rather than assigned success or failure.

Content and identifiers. An exploratory content-separation experiment used the same preparation requirement and environment evaluator. If s_e^{id} preserves the identifier subset of the agent’s memory, its descriptive contrast is

$$\Delta_{\text{content|id}} = \mathbb{E}_e \left[\Pr(Y_e(s_e) = 1) - \Pr(Y_e(s_e^{\text{id}}) = 1) \right]. \quad (6)$$

The saved goal and unrelated record are unchanged. Each configuration contributes 16 tasks, two per mechanism, with two continuations in each of six conditions. Paired tasks are re-sampled within mechanism for 5,000 descriptive draws (seed 2026092602). The study was designed after inspecting the whole-memory intervention and is analyzed as development evidence. All planned tasks and preparation failures are retained. These development intervals and supplementary specificity tests are reported separately from the original primary family.

The content verification used 128 new memory tasks per API configuration, with 16 tasks per mechanism and two continuations under each of eight conditions. Identity-only and content-only interventions partitioned the actual written memory. Exact JSON text and value-preserving rewording separated the effects of content, container and wording. The primary paired success differences used 10,000 task-bootstrap draws stratified by mechanism; 50,000 task-sign-flip draws and Holm adjustment evaluated the two model-specific primary contrasts. All 4,096 planned continuations entered the primary denominators. Supplementary Methods give the field partitions, secondary comparisons, encoding-equivalence analysis and complete acquisition accounting.

5.3. Predictive inference and budgets

Information available to the model. The shared repository context is retrieved from the visible issue text using lexical path scores, with deterministic path-order tie breaking. All arms receive the same fifty candidate paths and eight source excerpts; the exact tokenization, file filters, ranking formula and excerpt limits are given in Supplementary Methods, *Shared repository retrieval*.

The shared baseline uses seven numeric features: issue-text length in characters, retrieved-path count, mean pairwise agreement among the three base answers, normalized file-vote margin, file-vote entropy, invalid-answer fraction and mean reported confidence. Model and repository identities are categorical features. If $v_{(1)} \geq v_{(2)}$ are the two largest file-vote counts, the margin is $(v_{(1)} - v_{(2)})/3$, with an absent second count set to zero. Entropy is $-\sum_j p_j \log p_j$, where $p_j = v_j / \sum_\ell v_\ell$; both statistics are zero without valid file votes. Confidence averages reported values in $[0, 1]$ and is missing if none are valid. The following paragraphs define set agreement and training-fold preprocessing.

Linear diagnostic representation. In the fixed linear comparison, each strategy receives the same visible issue/code information and the same original working state, with three shared calls plus five strategy-specific calls. The state probes are current state, minimal candidate-file state, removed state, an irrelevant note and a synthetic mismatch drawn from the same visible candidate pool. Generic candidates were five additional self-consistency calls and five instruction paraphrases. Each augmentation supplies seven features to an equally specified logistic predictor. The prespecified development rule selected self-consistency before formal collection: paraphrase improved its own grouped development AUC by 0.008, below the 0.020 threshold for switching. This selection did not compare either candidate against CSB.

Let $J(A, B) = |A \cap B| / |A \cup B|$, with empty or invalid file sets assigned similarity zero, and let C be the top-three file-vote consensus from the base calls. Writing $P_c, P_m, P_{\text{removed}}, P_i, P_x$ for the current, minimal, removed, irrelevant and mismatched-state output sets, the first five state features are $J(P_c, C)$, $J(P_c, P_m)$, $1 - J(P_c, P_{\text{removed}})$, $J(P_c, P_i)$ and $1 - J(P_c, P_x)$. The two difference features are zero if either output is invalid. The remaining two features are the invalid-output fraction and mean reported confidence. Generic features use mean agreement with C , pairwise agreement, normalized vote margin, path-vote entropy in nats, mean base-to-extra agreement, invalid fraction and mean confidence. Unavailable confidence values receive training-fold median imputation.

Linear prediction and grouped validation. For the fixed linear comparison, all preprocessing is fitted inside five-fold grouped training splits (shuffled group identifiers, seed 17). Numeric features receive training-fold median imputation and standardization; model and repository use one-hot encoding with unknown categories ignored. With $\phi(x)$ denoting training-fold-transformed features, the fitted probability and incremental discrimination are

$$\hat{p}(y = 1 \mid x) = \{1 + \exp[-\beta_0 - \beta^\top \phi(x)]\}^{-1}, \quad \delta = \text{AUC}_{\text{CSB}} - \text{AUC}_{\text{generic}}. \quad (7)$$

The probability predictor is L2 logistic regression with $C = 1$, the liblinear solver, 2,000 maximum iterations, seed 17 and no class weighting. All rows for one issue remain together; visible issue near-duplicates or identical repository/base-commit pairs are additionally grouped. In this fixed linear comparison, state and generic diagnostics use the same

feature count and estimator specification. The nonlinear comparison gives both arms the same representation choices, learner families and search budget; their selected representations can differ. Bridge confidence intervals repeat the grouped fit and preprocessing in 2,000 issue/template-group bootstrap draws, keeping duplicate copies of the same group within one fold. Non-estimable draws are counted. Precision-recall discrimination, Brier score, per-model results and repository- and model-held-out results complement pooled AUC. A single-class test slice is marked non-estimable. For the 100-issue reassessment, the test-cluster bootstrap conditions on predictors refitted and frozen on the four-model development cohort.

Cohorts and evaluation roles. The development cohort contains 300 SWE-bench Lite test issues, also used by the six-model reference-dependent analysis and termed the bridge in the supplementary figures. Twenty separate Lite development issues were used to check the implementation and select the generic comparator. Astra uses a fixed 60-issue development subset. The 100-issue SWE-bench Verified reassessment excludes duplicate IDs, repository/commit matches and prespecified text/patch near-duplicates.

The 100-issue outcomes had been observed before the model-cohort choice, with labels excluded from fitting and selection; this evaluation therefore conditions on that choice (Dwork et al., 2015). Supplementary Note 14 gives the service-driven substitution of DeepSeek for Grok and the preceding cohorts. The independent 800-issue sample was unused in development and acquired after predictor selection, from the full SWE-bench test population rather than the Lite or Verified subsets. Supplementary Methods and the public membership lists specify the cohort composition and exclusion rules.

Inference-budget comparisons. The generic diagnostic starts with five queries and can continue to ten until its reported tokens reach the state-probe cost. For the independent evaluation, it must reach the larger of the full-state and repetition-intervention costs. Stopping depends only on token usage, so the final indivisible query can overshoot the target. When metering is complete and the target is reached, the comparator receives at least as many diagnostic queries and reported tokens as the candidate. The shared three base calls are charged once in joint acquisition and to each strategy in deployment accounting. Costs include transport attempts and reported input, output and reasoning tokens. All planned records, including terminal failures, enter the primary comparison; same-call and complete-and-funded comparisons are reported separately. Retrospective prefix analyses include the acquired over-budget query and identify unmetered responses separately. Supplementary Notes 13–15 provide the detailed accounting for the linear comparisons.

Richer representations and final predictors. The richer representation describes three aspects of a diagnostic response: its quality as a localization answer, its agreement with other answers, and its correspondence with the supplied state. Each response contributes 21 reference-free quantities. Five concern confidence, validity, parsing success, path count and valid-path fraction; four compare it with the base responses; five compare state and answer. The remaining seven measure retrieval coverage, best and mean retrieval rank, path-query and returned-state-query lexical overlap, serialized state length and whether that state is nonempty.

File paths are normalized, checked against the visible repository and limited to three implementation files. Set agreement is $J(A, B) = |A \cap B|/|A \cup B|$ for two nonempty sets

and zero otherwise. Base-response features compare the returned set with the first base answer and the three most frequent base paths, and retain rank-weighted votes and top-path agreement. State features compare it with the input, returned and original candidate-file sets, retain its change in input-state agreement relative to the first base answer, and measure the input state’s distance from the original state. An empty input state has missing distance. Retrieval ranks range from zero to 49 among the visible candidates, with absent paths assigned 50; best and mean ranks are divided by 50 and equal one for an empty output. Lexical overlap uses lower-cased identifier tokens after stop-word removal. Supplementary Methods give the vote normalization, tokenization and the full list of quantities.

We summarize these quantities in two ways. The aggregate representation describes their distribution across responses using finite-value mean, population standard deviation, minimum, maximum and missing fraction. It also records pairwise file-set overlaps, the proportion of unique file sets, file-vote entropy in nats, the maximum vote count divided by the number of calls, and that call count. All funded generic calls enter these summaries.

The ordered representation retains which of the first five diagnostic responses produced each value. It contains their five response vectors, ten pairwise overlaps and valid-response versions, and differences in confidence and state/base agreement from the first response. A valid-response pair is missing unless both outputs pass validation; absent confidence remains missing. Both representations accompany the common base features and seven original diagnostic features. Table 6 reports the selected representations. Supplementary Methods and the frozen model records specify the complete ordered contrasts and column lists.

Training-fold preprocessing prevents validation observations from influencing the fitted transformations. Numeric features are converted to float32, then missing values receive the training-fold median. Entirely missing columns are retained with zero values; indicators are added for columns containing missing training values. Numeric columns and their indicators are standardized. Model and repository identities use one-hot encoding, with unseen categories ignored.

Each diagnostic strategy receives the same four representation choices and 60 candidate configurations, evaluated in five outer issue/template-group folds with three inner folds. For each outer validation fold, inner out-of-fold AUC selects one member of each learner family. Final models use each family’s configuration with the best pooled five-fold out-of-fold development AUC. Both selections break ties by log loss and then by a fixed candidate order. The selected estimators are refitted on all 300 development issues, and their probabilities are averaged with weight 0.25 each. These predictors are fixed before the 800 new issues are acquired (Cawley and Talbot, 2010).

Table 6: Final configurations for the primary independent comparison. Aggregate and ordered-plus-aggregate representations contain 130 and 279 columns before preprocessing and categorical expansion. Each learner contributes one quarter of the predicted probability.

Learner	Repetition–intervention	Token-funded generic
CatBoost	Aggregate; depth 4; L2 penalty 3	Aggregate; depth 6; L2 penalty 10
Extra trees	Aggregate; depth 8; minimum leaf 5	Ordered + aggregate; depth 8; minimum leaf 5
Histogram boosting	Aggregate; maximum 15 leaves; minimum leaf 10	Aggregate; maximum 15 leaves; minimum leaf 10
Logistic regression	Aggregate; $C = 0.1$	Aggregate; $C = 0.1$

CatBoost uses 160 iterations, learning rate 0.05 and log loss. Extra trees uses 160 trees and all features at each split. Histogram boosting uses 150 iterations, learning rate 0.05, L2 penalty 5 and no early stopping. Logistic regression uses the liblinear solver with at most 3,000 iterations. Estimator random seeds are 17. The early linear comparison retains its separate $C = 1$ specification.

Independent predictive evaluation. The primary independent comparison is repetition–intervention versus token-funded generic diagnosis. Pure state interventions and the frozen noise-contrast variant form two prespecified secondary AUC comparisons, each against its corresponding generic predictor. Ten thousand paired issue-bootstrap draws stratified by repository carry all four model records together and leave the trained estimators fixed. We report marginal 95% intervals and 98.333% intervals for the three-comparison Bonferroni family. Brier score, log loss, calibration in fixed probability bins and failure rates at fixed retained fractions are prespecified secondary descriptions. These intervals quantify new-issue variation conditional on training and the twelve sampled repositories; no new outcomes enter fitting, threshold selection, model replacement or sample-size decisions.

5.4. Sampling, model-visible information and annotation

Each model in the controlled open-weight study contributes 126,000 scored rows: three scaffolds, 20 task families, 100 events per family, seven variants and three replicate indices. Of these, 108,000 are model-generation records and 18,000 are offline-policy records. The offline policy uses the frozen expected-action marginal within selected events; it is not an empirical prior fitted to structured-model outputs. The seven variants include post-hoc/scrambled control rows. The strict decisive-field lesion is a separate additional slice, not one of the seven core variants.

The seven source corpora support distinct baseline and finite-action samples. The baseline design samples up to 500 events per corpus under eight conditions and three random seeds. The complete baseline acquisition contains 3,098 entries (3,073 unique corpus–event IDs) and all 74,352 planned responses. The finite-action sample contains 818 entries and 803 unique identifiers, from six corpora with 120 entries each and Narratives with 98. It shares 73 identifiers with the 224-event source intake. Supplementary Note 2 gives the full source-membership relationships, including the separate 497-entry snapshot, configured caps, observed counts and generation totals. The baseline replication reports raw-field coverage and reason-response contrasts separately from compatibility scores calculated using rule-completed fields.

The human evaluation sampled 1,200 items from 532,800 candidate records in the Qwen2.5-7B, 14B and 32B and Mistral-7B experiments and their controls, selecting 150 items per condition. Three graduate students in relevant disciplines, including doctoral and master’s students, independently rated every item using a common seven-dimension codebook and written instructions. Participation was voluntary, and annotators were informed that their ratings would be used for this research. The codebook specified binary judgment criteria and handling of unjudgeable cases. Annotators received the same masked items in the same randomized order and scored only visible information. Supplementary Note 5 details source-stratified sampling, masking, score aggregation and agreement analysis.

Supplementary Methods Tables M1–M3 map each sample to its analysis role and specify what intervention and predictor information is available.

5.5. Reproducibility and analysis specification

The public research archive contains the task-level observations, complete prompts, model and inference specifications, environment snapshots and analysis inputs needed to reproduce the reported comparisons. Cohort lists and analysis specifications identify the prespecified tests and exploratory analyses. Request records and source hashes link these materials to the full planned samples and plotted summaries.

6. Discussion

Agent-written state contributes to action through the task information it carries. The controlled experiments establish consistent responses to supplied information; the execution experiments identify dependence on records produced and retained by the acting system. In Sol and DeepSeek, removing memory or a persistent goal reduces transaction success, restoration recovers performance, and irrelevant perturbations have much smaller effects. Preserving transaction identifiers while removing other memory content leaves a large pooled deletion effect. All eight mechanism-specific point estimates are positive, with varying magnitudes. The evidence connects a realized record to executable choices and shows which parts of that record matter.

The information-access controls locate this dependence within the broader problem of memory utilization. Mem2ActBench evaluates whether remembered facts can be assembled into tool parameters; its main comparison supplies the target tool to focus on parameter grounding (Shen et al., 2026). Comparisons of memory-writing and retrieval strategies further show how failures to access stored information can be separated from failures to apply it (Yuan et al., 2026). Here, the fixed checkpoint holds the agent’s actual written record constant. Restating relevant facts at handoff or reopening earlier history compensates for memory removal, while restoring the record recovers execution. Dependence therefore reflects the routes by which information reaches an action: the same memory can matter strongly when it is the sole source and little when another source supplies the facts. A deletion effect is consequently a property of the record within its available context and continuation policy. This explains why a small effect in a redundant context can coexist with effective use of the same information.

Content replacement identifies what the record leads the agent to do. Actions move toward the target specified by the altered record, even when that target departs from the

original task requirement. Interchange interventions on neural representations use this directional criterion to test a proposed correspondence between a representation and a causal variable (Geiger et al., 2021); mediation studies likewise compare answers under altered intermediate reasoning while holding the problem fixed (Paul et al., 2024). In the present experiments, the manipulated variable is an exposed record, and the response is an executable transaction. Scoring that transaction against both targets separates following the replacement from an undirected loss of success. Together with deletion and compensation, the comparison distinguishes lack of access, dependence on accessible content, and consistent action on an inaccurate record. State use and record correctness are thus separate properties with different consequences for task completion.

Encoding also affects how information is used. The content experiment compares a structured record, its exact JSON serialization and value-preserving rewording, so the effect of removing facts can be interpreted alongside changes in presentation. Sol’s difference between the two text conditions shows sensitivity to encoding despite identical task values. The two text conditions met the prespecified descriptive equivalence criterion in DeepSeek. This model dependence accords with intervention studies in which reliance on intermediate text varies across tasks and configurations (Lanham et al., 2023).

The software application asks whether these response relations help predict localization success. SelfCheckGPT uses agreement among sampled responses to assess factuality (Manakul et al., 2023), while semantic entropy measures variation among meanings rather than word sequences (Farquhar et al., 2024). These results explain why additional generic observations are a substantive comparator. Semantic-entropy detection also distinguishes variable answers from systematic errors that remain consistent across samples (Farquhar et al., 2024). Our content-replacement results illustrate the corresponding behavioral problem: an agent can use a record consistently and execute the wrong transaction. Intervention sensitivity and repeated-answer agreement therefore provide different descriptions of behavior, whose predictive combination must be evaluated against correctness.

On the independent software sample, the repetition–intervention diagnostic produced an AUC point increment of 0.00382 and a Brier-score reduction of 0.00293 relative to token-funded generic queries. The matched comparison quantifies the predictive contribution of response relations beyond information already available from repeated queries. The observed increment was small and varied by model. Strong behavioral dependence can coexist with a small predictive increment when other observations already discriminate successful from unsuccessful outputs.

The eight synthetic mechanisms permit close control over task information and transaction correctness within short tool interactions. Four continuations per core condition and two per content-verification condition give discrete task-level success estimates; the content study includes 16 new tasks per mechanism. Low baseline success limits the resolution of state-dependence tests for the quantized open-weight configurations. The software results concern new issues from twelve repositories evaluated by fixed predictors. Longer interactions would test whether the same dependence persists as records accumulate, conflict or become obsolete. Comparing repair policies with equal information would address how identifying a consequential record can improve task completion.

7. Conclusion

Structured state supports consistent controlled action responses, and agent-written memories and goals make substantial, recoverable contributions to tool execution in the tested API configurations. Specificity, restoration and alternative information access show when a saved record matters. Content separation and replacement distinguish the use of task information from the need for identifiers and from the correctness of the resulting action. On new software issues, the hybrid diagnostic yielded a small pooled AUC point increment and lower Brier loss relative to token-funded generic queries. Together, the experiments connect the information an agent retains to the actions it subsequently executes, providing an empirical account of state-dependent action.

Data and code availability

Analysis-ready data and code supporting the findings are publicly available at <https://github.com/JiaXiao-Peter/Causal-State-Binding>. The repository provides study protocols, cohort membership lists, task-level outcomes, frozen localization predictions and estimators, scoring and analysis code, and source data for all figures. The versioned research package includes commands to reproduce the reported summaries and figures from saved observations. Third-party benchmark materials retain their original licenses and access terms.

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CRedit authorship contribution statement

Xiao Jia: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing - original draft, Writing - review and editing, Visualization.

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- Cawley, G.C., Talbot, N.L.C., 2010. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research* 11, 2079–2107. URL: <https://www.jmlr.org/papers/v11/cawley10a.html>.
- Dwork, C., Feldman, V., Hardt, M., Pitassi, T., Reingold, O., Roth, A., 2015. Generalization in adaptive data analysis and holdout reuse. URL: <https://arxiv.org/abs/1506.02629>, doi:10.48550/arXiv.1506.02629, arXiv:1506.02629.
- Farquhar, S., Kossen, J., Kuhn, L., Gal, Y., 2024. Detecting hallucinations in large language models using semantic entropy. *Nature* 630, 625–630. URL: <https://www.nature.com/articles/s41586-024-07421-0>, doi:10.1038/s41586-024-07421-0.
- Geiger, A., Lu, H., Icard, T., Potts, C., 2021. Causal abstractions of neural networks, in: *Advances in Neural Information Processing Systems*. URL: <https://papers.nips.cc/paper/2021/hash/4f5c422f4d49a5a807eda27434231040-Abstract.html>.
- He, J., Shi, M., Jia, J., Yang, X., Sun, D., 2026. What process evaluation of coding agents actually measures: Action, task, and step are three different levels. arXiv:2608.22960v1. URL: <https://arxiv.org/abs/2608.22960v1>, doi:10.48550/arXiv.2608.22960.
- Jimenez, C.E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., Narasimhan, K.R., 2024. SWE-bench: Can language models resolve real-world github issues?, in: *The Twelfth International Conference on Learning Representations*. URL: <https://openreview.net/forum?id=VTF8yNQM66>.
- Kuhn, L., Gal, Y., Farquhar, S., 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation, in: *International Conference on Learning Representations*. URL: <https://openreview.net/forum?id=VD-AYtP0dve>.
- Lanham, T., Chen, A., Radhakrishnan, A., Steiner, B., Denison, C., Hernandez, D., et al., 2023. Measuring faithfulness in chain-of-thought reasoning. arXiv preprint arXiv:2307.13702 URL: <https://arxiv.org/abs/2307.13702>.
- Liu, X., Yu, H., Zhang, H., Xu, Y., Lei, X., Lai, H., Gu, Y., Ding, H., Men, K., Yang, K., Zhang, S., Deng, X., Zeng, A., Du, Z., Zhang, C., Shen, S., Zhang, T., Su, Y., Sun, H., Huang, M., Dong, Y., Tang, J., 2024. AgentBench: Evaluating LLMs as agents, in: *The Twelfth International Conference on Learning Representations*. URL: <https://openreview.net/forum?id=zAdUB0aCTQ>.
- Lu, J., Holleis, T., Zhang, Y., Aumayer, B., Nan, F., Bai, H., Ma, S., Ma, S., Li, M., Yin, G., Wang, Z., Pang, R., 2025. ToolSandbox: A stateful, conversational, interactive evaluation benchmark for LLM tool use capabilities, in: *Findings of the Association for Computational Linguistics: NAACL 2025*, Association for Computational Linguistics, Albuquerque, New Mexico. pp. 1160–1183. URL: <https://aclanthology.org/2025.findings-naacl.65/>, doi:10.18653/v1/2025.findings-naacl.65.

- Manakul, P., Liusie, A., Gales, M., 2023. SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models, in: Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics. pp. 9004–9017. URL: <https://aclanthology.org/2023.emnlp-main.557/>, doi:10.18653/v1/2023.emnlp-main.557.
- Park, J.S., O’Brien, J.C., Cai, C.J., Morris, M.R., Liang, P., Bernstein, M.S., 2023. Generative agents: Interactive simulacra of human behavior, in: Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology. doi:10.1145/3586183.3606763.
- Paul, D., West, R., Bosselut, A., Faltings, B., 2024. Making reasoning matter: Measuring and improving faithfulness of chain-of-thought reasoning, in: Findings of the Association for Computational Linguistics: EMNLP 2024, Association for Computational Linguistics, Miami, Florida, USA. pp. 15012–15032. URL: <https://aclanthology.org/2024.findings-emnlp.882/>, doi:10.18653/v1/2024.findings-emnlp.882.
- Shah, J., 2026. Causal agent replay: Counterfactual attribution for LLM-agent failures. arXiv:2606.08275v1. URL: <https://arxiv.org/abs/2606.08275v1>, doi:10.48550/arXiv.2606.08275.
- Shen, Y., Li, K., Zhou, W., Hu, S., 2026. Mem2ActBench: A benchmark for evaluating long-term memory utilization in task-oriented autonomous agents, in: Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics. pp. 8173–8190. URL: <https://aclanthology.org/2026.acl-long.370/>, doi:10.18653/v1/2026.acl-long.370.
- Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., Yao, S., 2023. Reflexion: Language agents with verbal reinforcement learning, in: Advances in Neural Information Processing Systems. URL: https://papers.nips.cc/paper_files/paper/2023/hash/1b44b878bb782e6954cd888628510e90-Abstract-Conference.html.
- Turpin, M., Michael, J., Perez, E., Bowman, S.R., 2023. Language models don’t always say what they think: Unfaithful explanations in chain-of-thought prompting, in: Advances in Neural Information Processing Systems, pp. 74952–74965. URL: https://proceedings.neurips.cc/paper_files/paper/2023/hash/ed3fea9033a80fea1376299fa7863f4a-Abstract-Conference.html.
- Wang, X., Wei, J., Schuurmans, D., Le, Q.V., Chi, E.H., Narang, S., Chowdhery, A., Zhou, D., 2023. Self-consistency improves chain of thought reasoning in language models, in: International Conference on Learning Representations. URL: <https://openreview.net/forum?id=1PL1NIMMrw>.
- Xia, C.S., Deng, Y., Dunn, S., Zhang, L., 2024. Agentless: Demystifying LLM-based software engineering agents. arXiv:2407.01489v2. URL: <https://arxiv.org/abs/2407.01489v2>, doi:10.48550/arXiv.2407.01489.

- Yao, S., Shinn, N., Razavi, P., Narasimhan, K.R., 2025. τ -bench: A benchmark for tool-agent-user interaction in real-world domains, in: The Thirteenth International Conference on Learning Representations. URL: <https://openreview.net/forum?id=roNSXZpUDN>.
- Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., Cao, Y., 2023. ReAct: Synergizing reasoning and acting in language models, in: International Conference on Learning Representations. URL: https://openreview.net/forum?id=WE_vluYUL-X.
- Yuan, B., Su, Y., Yao, K., 2026. Diagnosing retrieval vs. utilization bottlenecks in LLM agent memory. arXiv:2603.02473v1. URL: <https://arxiv.org/abs/2603.02473v1>, doi:10.48550/arXiv.2603.02473.
- Zhang, H., 2026. Credit without ground truth: Auditing step-level credit assignment in LLM agents against executed replay. arXiv:2608.19760v1. URL: <https://arxiv.org/abs/2608.19760v1>, doi:10.48550/arXiv.2608.19760.
- Zhou, S., Xu, F.F., Zhu, H., Zhou, X., Lo, R., Sridhar, A., Cheng, X., Ou, T., Bisk, Y., Fried, D., Alon, U., Neubig, G., 2024. WebArena: A realistic web environment for building autonomous agents, in: The Twelfth International Conference on Learning Representations. URL: <https://openreview.net/forum?id=oKn9c6ytLx>.